

Homework 1

Total points : 14

1. Generate 500 observations from the standard normal distribution and draw a histogram based on this data. On the histogram, plot kernel density estimates based on the same data for various choices of well-known kernels. Compare the performance among the Histogram estimators and various kernel density estimators.

Points : 7

2. Generate 100 observations (y_i, x_i) ($i = 1, \dots, 100$) on the model $Y = e^X + \epsilon$, where X follows uniform distribution over $(0, 1)$, and $E(\epsilon|X) = 0$ almost surely. Compare the performances of Nadaraya-Watson Estimators when ϵ follows standard normal and standard Laplace distributions.

Points : 7

Deadline: Deadline of submission is 10 PM, January 26, 2026.

Remark: You can use any package or any programming language to carry out the study.